

Project Report

Project Title: Stack Applications in C++

1. Introduction

Stack is a linear data structure that follows the Last In First Out (LIFO) principle. This project demonstrates stack implementation and its practical applications using C++.

2. Objectives

- Understand stack operations.
- Implement stacks using arrays and linked lists.
- Apply stacks to solve expression problems.
- Simulate browser navigation using stacks.

3. Technologies Used

- Language: C++
- IDE: Visual Studio Code
- Compiler: g++
- Operating System: Linux (Ubuntu)

4. Modules Implemented

1. Stack using Array
2. Stack using Linked List
3. Parentheses Matching
4. Infix to Postfix Conversion
5. Postfix Expression Evaluation
6. Browser History Simulator

5. Algorithms

Array Stack: Push, Pop, Peek, Display using fixed-size array.

Linked List Stack: Dynamic stack using linked list nodes.

Parentheses Matching: Push opening brackets and validate closing brackets.

Infix to Postfix: Uses operator precedence and a character stack.

Postfix Evaluation: Uses an integer stack to evaluate postfix expressions.

Browser History: Uses two stacks (Back and Forward) to simulate browser navigation.

6. Features

- Menu-driven application
- Modular source files
- Input validation
- Real-world stack applications
- Clean project structure

7. Time Complexity

Push: $O(1)$

Pop: $O(1)$

Peek: $O(1)$

Display: $O(n)$

Parentheses Matching: $O(n)$

Infix to Postfix: $O(n)$

Postfix Evaluation: $O(n)$

8. Results

The application successfully performs stack operations and demonstrates multiple practical applications of the stack data structure through an interactive console interface.

9. Conclusion

This project strengthened understanding of stack data structures, modular programming, and algorithm design in C++. It demonstrates both theoretical concepts and practical applications.

10. Future Enhancements

- Support multi-digit postfix evaluation
- Template-based generic stack
- File-based browser history
- GUI implementation using Qt or similar framework

References

1. Data Structures using C++ textbooks
2. C++ Standard Library Documentation
3. Course materials and lecture notes